SUBRAHMANYA BHAT, PMP
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An experienced Engineering Professional with Project Management Professional (PMP) expertise worked on Techno Managerial roles in embedded systems from Avionics, Consumer Electronics to Automotive including Infotainment & Telematics (Connected Cars) systems development, integration, E2E validation and Product launches.

- Technology leader who possesses UNIQUE competencies in Automotive software development/ system integration/E2E system validation of Avionics, Consumer Electronics and Automotive embedded system projects Global Release and Program management.
- As a Project Management Professional (PMP) since 2008, I bring extensive experience in successfully executing complex projects with cross-functional teams, prioritizing, collaborating, and delivering high-quality results to ensure customer satisfaction.
- Expertise in managing the entire project development life cycle, ensuring alignment with cost, time, and quality objectives.
- From project inception to closure, I have been responsible for all aspects, including initiation, planning, development, implementation, maintenance, and closure.
- Developed and maintained a roadmap for In-Vehicle Infotainment features, aligning product development team with market demands.
- Possess a keen eye for detail and is extremely capable of multitasking in a fiercely competitive and fast-paced environment. Excellent in identifying and building strategic and long-term business relationships with senior key stakeholders.
- Proven experience in leveraging technical leadership skills to deliver complex programs/projects.
- Supported product engineering team, providing system engineering specific knowledge for out- sourced programs, and in-house Telematics & Infotainment projects at General Motors.
- Developed and maintained the system engineering plan for the project according to the GM System Engineering Process for assigned projects.
- Possess Team development and Leadership skills with Program Management and Execution.
- Led engineering development activities in multiple geographical regions (NA, Mexico, China & Europe) for successful product delivery.
- Experienced professional in System Engineering process with special focus on requirements and software engineering processes.
- Expertise in In-Vehicle Infotainment engineering, Telematics/Connected cars, vehicle integration, electrical architecture, Telematics Change Control and related other domains.
- Expertise in developing In-Vehicle Infotainment (IVI) app software, along with troubleshooting skills in Android automotive systems. My experience includes offboard communication using MQTT and PubSub models, leveraging Kubernetes microservices for both Internal Combustion Engine (ICE) and Battery Electric Vehicle (BEV) vehicle lines.
- Well-versed with General Motors - Global Vehicle Development Process (GVDP) and functional safety ASIL standards for GM Telematics (OnStar) module and compliance
- Hands on software development experience in C for mission critical and safety critical avionics projects.
- Strong understanding of the Electrical Architectures of the GM like Global A and Global B and Stellantis NEA, Atlantis high, Mid and VP4.
- Expertise in focus areas like Infotainment & Telematics, In-Vehicle Apps integration, Diagnostics, Vehicle Data upload, Vehicle network data logging, Vehicle to Edge & Edge to Cloud data communications, Embedded navigation etc.
- Experience in updating Program status in Core PDT meetings to leadership forum, development and validation teams to ensure schedule aligns the software, hardware and components releases to GVDP and VeSCoM.
- Expertise in understanding of functional safety ISO 26262 for Automotive ECUs.

- Expertise in Avionics RTCA/ DO-178B Software Considerations in Airborne Systems and Equipment Certification

TECHNICAL QUALIFICATION

- Languages: C, C++, Java, ADA, Assembly Language Programming.
- Software Tools: NEO-VI, Global Sim, DPS, VSpy, Intrepid Tools, Vector CANalyzer/CANCaseXL. Wireshark, QXDM, Putty, IBM Rational DOORS, IBM Rational DOORS Next Generation (DNG), RTC, Polarion, Jira, PRTS, ESIM database.
- Protocols/Standards: CAN, CAN FD, LIN, MOST, DTCs, Ethernet Standard/Automotive, FlexRay, LIN, MOST, GMLAN, Global A, Global B, RS232, RS422, MIL-STD-1553B, DO-178B, ISO 26262, DOD 2167A, A-SPICE, ASIL Standards, DIDs
- Operating Systems: UNIX, Linux, Microsoft Windows

WORK EXPERIENCE

Senior Technical Product Manager/Senior Systems Technical Lead – Connected Vehicle.
Stellantis (Former FCA), Auburn Hills, MI July 2021 – 03/31/2024

- Leading a Global cross-functional team of system engineers, developers, back-office experts and suppliers to implement the new Connectivity features in the Infotainment & Connectivity platform.
- My Team is responsible for the timely delivery of the Connectivity product features and its final quality by collaborating with business development departments to understand their requirements and by ensuring the highest value is delivered.
- Successfully used Agile Scrum to manage Dynamic Pricing project providing modern price management capability increasing net income.
- Serve as a Sr. Transformation Coach across Retail, Preferred Rewards, Small Business, Wealth
- Coach new teams in transforming to Lean Agile using SAFe 5.1 and SAFe 6.0 (Lean, Scrum, Kanban and SAFe).
- Assess the current organization, team and delivery environment and recommend optimal Agile practices
- Took full ownership of the entire product lifecycle, from initial discovery to final delivery, collaborating with diverse stakeholders. This entails overseeing all aspects of design, development, marketing, safety, and legal compliance to mitigate risks and ensure value, usability, viability, and feasibility throughout the process.
- Responsible for building and maintaining a roadmap for the development of In-Vehicle Infotainment features.
- As an Agile Coach/Practitioner, established and drove Agile methodologies and Release Planning processes for development teams across multiple Geographies (US/Overseas).
- Responsible to make sure that the design, architecture, and lower-level requirements are reviewed, and robust design and development plan is elicited, and software released on time.
- I am accountable for driving deliverables through all cross-functional teams, including Onboard, Offboard, Human-Machine Interface (HMI), Inter-Process Communication (IPC), and Software Factory. I ensure that software for the product is delivered according to the vehicle development timeline.
- Help ensuring that all dependencies are satisfied and removing blocking points.
- Responsible to build a strong technical background to understand to a quite fine level of detail the mechanisms and issues that will impact the feature.

Senior Manager – Systems Engineering.
Aptiv LLC, MI May 2019 – July, 2021

- Led a cross-functional team of system engineers, developers and data engineers to develop an Active Safety data collection and analytics platform focused on improving active safety product development

efficiencies. This also enables to gain customer insights to enhance safety on autonomous and/or semi-autonomous pathways.

- Provided role-based Scaled Agile Framework training for executives (business and technology), Product Owners, Scrum Masters, Release Train Engineers (RTE), and system teams
- Managed a team of System Engineers and Software Developers on Agile methodology to deliver the project to our key OEMs including the prioritization of product backlog, Sprint planning, Sprint reviews, monitoring & resolving the roadblocks and deploying the quality software on production.
- Leading & driving the System Engineering – Requirements & Software Engineering aspects of the Project. This is an IoT based project with Cloud and Edge computing. Tools used are Polarion, requirements -test cases traceability via Jira tool.
- Coach multiple Scrum Masters and their teams in Lean Agile principles and Scrum practices to optimize benefits of being Agile.
- Coach new teams in transforming to Lean Agile using SAFe4.5 (Lean, Scrum, Kanban and Safe).
- Involved in the Automotive SPICE assessments & compliance for the project.
- Responsible for Product Change management from gathering Stakeholder requirements, generating the System Requirements, System test cases, Vehicle Integration testing to final deployment.
- Technical review of design artifacts, System & software requirements to ensure robust design. Ensure the proper DFMEA is carried out on the design & requirements.
- Hands-on skills to write detailed user stories and robust system requirements.
- Managed Active Safety (ADAS) Realtime data logging application development, Integration and deployment which collects data from various Vehicle networks (CAN, CAN FD, Ethernet etc.)

Lead Design Release Engineer (DRE) – Telematics Engineering
General Motors, MI Oct 2018 – May 2019

- Worked collaboratively between systems, product development, validation, Systems Integration, and Backoffice teams owning the complete development plan which aligns all software development, validation, back office, integration, and other feature readiness tasks to align with GVDP timelines.
- Led the development of ALDL (Diagnostics spec) for Gen11 Global A and Global B Telematics Modules. Major test bench setups for Gen11 all variants.
- Executed Senior DRE responsibilities for the Telematics ECU for Global A & Global B for Global Regions and delivered the parts in-line with GVDP timelines. Prioritization of PRTS issues for the program delivery.
- Support the team and enable Scrum Master, Product Owner and other roles to effectively perform their roles
- Coach teams through Agile routines and ceremonies including Iteration / Sprint Planning, Scrum, Backlog
- Assisted 3 teams in Agile Release Planning; Provided RTE/Program Manager guidance in creating Vision and Roadmap artifacts; create/update Release/Program Plans with features and dependencies.
- Actively facilitated the Gen11 PDT meetings with TCP Suppliers and validation leads.
- As PDT lead reviewed the DV & PV ADVP& R for the approvals.
- Led Telematics PDT for Gen11 Global B Programs, EWO approvals and follow-up.
- Led the team driving process and tool improvements to enhance operational efficiency.
- Collaborated closely with software teams to clarify rule-sets and eliminate ambiguity, providing guidance on program documentation interpretation from an option content perspective.
- I spearheaded formal design, peer, and quality reviews, while supporting the resolution of technical issues and facilitating product development team meetings.
- My role also involved releasing Telematics ECU part numbers, communicating effectively with internal and external stakeholders, and planning and implementing test and development programs.
- Lastly, I remained proactive in staying abreast of emerging technologies and competitive products, driving design for manufacturing initiatives, and collaborating across departments to achieve organizational objectives.

Telematics Change Management Lead

General Motors, MI Aug 2016 – Oct 2018

- Led Telematics TRB and combined Infotainment & Telematics CCB meetings.
- Handled all Telematics Change Requests in TRB discussions and pre-screened CRs are put in Agenda for CCB discussion with Infotainment and Telematics Directors.
- Determined the timing alignment with program needs and Telematics workload perspective, and CR fields in Rational Team Concert (RTC) are complete and accurate per CR lifecycle phase.
- Increased efficiency/effectiveness of CCB by processing Requirement Architecture Development (RAD), Retime, and Rejection ballots offline.
- Develop late CR matrix in RTC life cycle and presented in weekly PDT meeting with Director to get the right priority for each change.

Lead Telematics Requirements Engineer
General Motors, MI July 2014 – Aug 2016

- Led a group of System Engineers to meet project timelines with quality and success.
- Experience in applying the requirements methodologies and management of deliverables in a complex multi-project environment for Global Programs like China, Europe, NA, South America & Mexico.
- I was involved in formulating the requirements for Super Cruise functionality for Gen10 VCP in both the NA and China regions. Additionally, I contributed to generating requirements for various other OnStar features such as OTA (Over the Air) Updates, Remote Reflash conditions, and Wireless specifications, among others.
- Coach multiple Scrum Masters and their teams in Lean Agile principles and Scrum practices to optimize benefits of being Agile.
- Skilled in coaching cross-functional team members (Dev, QA, BA), Product Owners, and Product Managers to achieve best practices.
- Ability to facilitate the Lean-Agile methodology, training sessions and activities.
- Expertise in Telematics engineering, vehicle integration, electrical architecture and related domains. Well-versed with IBM DOORS for requirements generation.
- Ensured the Requirements are well written and undergoes the DFMEA process and thorough impact analysis is performed to ensure robustness before these are provided to supplier for development.
- Led a group of System Engineers to deliver the OFS (OnStar Functional Specifications) for global regions (Europe/China/NA) – Gen10/Gen11
- Led the requirements migration from OFS to various GIS specs for TCP – Gen11.
- Train independent and sustainable Engineers in team

Senior Project Engineer- Senior Systems Architect (Contract)
General Motors, MI June 2013 – July 2014

- End to End Life Cycle Validation Engineer, for Gen10 Telematics Module (VCP) and involved in the whole gamut of E2E test cycle and Customer use case testing.
- Led integration and verification of software features for Infotainment, Telematics systems on multiple vehicle electrical architectures.
- Automated some of the feature validation using OATS Script.
- Executed testing of whole functionality of Gen10 Telematics module which gave me an understanding of the entire Telematics module.
- Involved in the validation and Issue triaging of various In-vehicle Apps of Infotainment system and embedded navigation.
- Involved in issue triaging, Analysis and getting to closure with Suppliers.

Project Manager
MphasiS Corporation, NY April 2004 – June 2013

- Worked for various major clients like Philips Corporation, Harman International, John Deere, State Farm Insurance and Ally Financials.
- My main experience is in domains like Embedded system development, integration and validation for Philips Consumer Electronic products, Harman International Navigation & Automotive products, and John Deere's embedded system development for Wireless Gateway- CAN based Realtime Event Generation and management.
- As Project Manager, involved in various roles for different clienteles with responsibilities not limited to reviewing and analyzing existing system structure to provide solutions to enhance and improve systems.
- Participated in technical collaboration meetings and provided the project management team with periodical reviews on implementation status as well as weekly reports on task plan for custom software implementations.
- Reviewed project specifications, flow charts, process diagrams and systems documentations.
- Participated in Software design discussions and provided right inputs to enhance the system.
- Developed, directed, and integrated system testing methods, programming and documentation.
- As Project Manager, recruited employees with right skillsets, mentoring, guiding through the process and ensured the team members deliver successfully.

Senior Software Engineer(Contract) - India
Aeronautical Development Agency – Dept. of Defense Research Organization July 2000 – Nov 2003

- Development, Integration and Testing of RS422 (Avionics Bus) based Flight Data Acquisition system (LRU) for fighter aircraft which monitors the Safety Critical flight parameters.
- Integration and Testing of MIL-STD-1553B based Mission Critical Data Acquisition system (LRU) for fighter aircraft which monitors mission critical parameters.
- Participated in code reviews, walkthroughs and software validation.
- Ensure Design, development and testing adhere to DOD 2167A standards to meet the ruggedness.
- Conducted flightworthy environmental testing (EMI/EMC, Vibration, Thermal) on various test labs for the LRU developed.
- Integration on flight simulators, Rigs and on the Real flight and validation.

Certifications
- DFSS Black Belt Certification from GMU.
- Certified Project Management Professional (PMP) from PMI.
- Design Release Engineer (DRE) Essentials Training by General Motors
- Certified Software Test Engineer (CSTE). (Certification from Quality Assurance Institute, Florida, USA)

EDUCATION
Bachelor of Engineering (Electronics) - Bangalore University, India. Institution: BMS College of Engineering, Bangalore, India - Year of graduation: 1994